

21-22-1 复变函数期末试卷 (A) 答案

一. 1. B 2. B 3. C 4. B 5. C

二. 1. -1 2. $e^{\frac{\pi}{2}-2k\pi}$, $k \in \mathbb{Z}$ 3. $1/e$ 4. $2v+1=0$ 5. 3 6. $\frac{1}{2}$, $-\frac{2\pi}{3}$ 7. $2\pi i$, $m=1$; 0 , $m \neq 1$

三. $\frac{z}{z+1} = \sqrt[4]{1} = e^{\frac{k\pi i}{2}}$, ($k=0, 1, 2, 3$)

$k=1$ 时, $z = \frac{-1+i}{2}$; $k=2$ 时, $z = -\frac{1}{2}$; $k=3$ 时, $z = \frac{-1-i}{2}$.

四. 由于 u 具有二阶连续偏导数, 且满足 Laplace 方程, 所以 u 是调和函数.

由 C.-R. 方程 $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 6xy$ 得 $u(x, y) = 3x^2y + \phi(y)$.

从而由 $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ 得 $\phi'(y) = -3y^2$, $\phi(y) = -y^3 + C$.

所以 $f(z) = 3x^2y - y^3 + C + i(-x^3 + 3xy^2) = -iz^3 + C$.

由 $f(0) = 1$ 得 $C = 1$. 故 $f(z) = -iz^3 + 1$.

五. 1. $f(z) = -\frac{1}{z-i} \left(\frac{1}{z}\right)' = -\frac{1}{i} \frac{1}{z-i} \left(\frac{1}{1+\frac{z-i}{i}}\right)'$
 $= -\frac{1}{i} \frac{1}{z-i} \left[\sum_{n=0}^{+\infty} \frac{(-1)^n}{i^n} (z-i)^n\right]' = \sum_{n=1}^{+\infty} \frac{(-1)^{n-1}n}{i^{n+1}} (z-i)^{n-2}$, $0 < |z-i| < 1$.

$$f(z) = -\frac{1}{z-i} \left(\frac{1}{z}\right)' = -\frac{1}{z-i} \left(\frac{1}{z-i} \frac{1}{1+\frac{i}{z-i}}\right)' = -\frac{1}{z-i} \left[\frac{1}{z-i} \sum_{n=0}^{+\infty} (-1)^n i^n (z-i)^{-n}\right]'$$

$$= \sum_{n=0}^{+\infty} (-1)^n (n+1) i^n (z-i)^{-n-3}, \quad 1 < |z-i| < +\infty.$$

2. $z_k = \frac{1}{k\pi + \frac{\pi}{2}} + 1$ ($k \in \mathbb{Z}$) 是 $f(z)$ 一级极点 (4分)

$\because z_k \rightarrow 1$ ($k \rightarrow \infty$), $\therefore z = 1$ 不是孤立奇点 (2分)

$\because f(z) \rightarrow 1$ ($z \rightarrow \infty$), $\therefore z = \infty$ 是可去奇点 (2分)

六. 1. 原积分 $= \frac{1}{2} 2\pi i (\text{Res}[f(z), ai] + \text{Res}[f(z), bi]) = \pi i \left[\frac{i}{(a^2 - b^2)2a} - \frac{i}{(a^2 - b^2)2b} \right] = \frac{\pi}{2ab(a+b)}$.

$$2. \text{原积分} = \frac{4}{i} \int_{|z|=1} \frac{z}{(z^2 + 4z + 1)^2} dz = \frac{4}{i} \int_{|z|=1} \frac{z}{[z - (-2 + \sqrt{3})]^2 [z - (-2 - \sqrt{3})]^2} dz$$

$$= \frac{4}{i} 2\pi i \text{Res}[f(z), -2 + \sqrt{3}] = \frac{4}{i} 2\pi i \frac{4}{(2\sqrt{3})^3} = \frac{4\pi}{3\sqrt{3}}$$

七. 由题意得 $f(z) = \frac{B}{z-x_0} + g(z)$, 其中 $g(z)$ 在 $|z-x_0| < R$ 内解析.

从而 $g(z)$ 在 $|z-x_0| \leq \rho$ 上有界.

于是 $\left| \int_{C_\rho} g(z) dz \right| \leq \pi M \rho \rightarrow 0$ ($\rho \rightarrow 0^+$).

$$\text{又 } \int_{C_\rho} \frac{B}{z-x_0} dz = B \int_\pi^0 \frac{i\rho e^{i\theta}}{\rho e^{i\theta}} d\theta = -B\pi i.$$

$$\text{故 } \lim_{\rho \rightarrow 0^+} \int_{C_\rho} f(z) dz = -B\pi i.$$